

Sean Luka Girgis

Production Engineer / SRE | Python, Oracle/SQL, Monitoring, RCA & Data Pipeline Reliability

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Professional Summary

Senior production engineering, SRE-adjacent, data reliability, and performance professional with 20+ years of enterprise technology experience across Python, SQL, Oracle, PL/SQL, production support, monitoring, incident response, root cause analysis, capacity planning, automation, data pipeline reliability, CI/CD concepts, and cloud data workflows. Strong background supporting regulated enterprise environments, troubleshooting production data issues, optimizing SQL/reporting layers, documenting runbooks, improving monitoring visibility, and translating operational signals into practical remediation. Best aligned to Production Engineer / SRE roles requiring Python/SQL/Oracle, monitoring, RCA, capacity planning, automation, data pipeline support, CI/CD collaboration, and operational reliability; Snowflake, Java/Spring Boot microservices, SAP FI/CO, ABAP, SAP Solution Manager, and deep Azure DevOps/Jenkins ownership are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported production monitoring, capacity, and data reliability workflows across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and large-scale infrastructure telemetry sources.
- Built Python/Pandas/PySpark and SQL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw performance signals into validated dashboards, reports, forecasting datasets, and executive insights.
- Performed production troubleshooting and RCA-style analysis by tracing source feeds, transformation logic, SQL behavior, metric behavior, missing or late data, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Developed validation and monitoring checks for row counts, nulls, unmatched joins, abnormal trend changes, freshness, and output consistency to improve pipeline reliability and reduce silent data issues.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Applied capacity planning and forecasting workflows using Prophet, scikit-learn, Python, SQL, and telemetry datasets to identify bottleneck risk and support proactive infrastructure planning.
- Created runbooks, metric definitions, validation checkpoints, support procedures, troubleshooting notes, and operational documentation to improve repeatability and incident response.
- Partnered with infrastructure, application, analytics, engineering, and business teams to prioritize issues, explain findings, coordinate fixes, and improve operational reliability.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed performance analysis and monitoring for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.

- Correlated load-test behavior with APM telemetry to identify bottlenecks, user-impacting symptoms, and improvement areas before production impact.
- Led the FAST project, mining real-user and synthetic monitoring metrics to identify performance improvement opportunities adopted by engineering teams.
- Created dashboards and reporting views that converted monitoring and test signals into actionable performance, reliability, and stakeholder insights.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined monitoring standards, dashboards, alert rules, SLA thresholds, validation checks, and operational reporting artifacts used by application, middleware, infrastructure, and operations teams.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable operational processes.
- Provided mentoring, onboarding support, troubleshooting guidance, and documentation to improve team adoption of monitoring and performance-analysis practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, NFR-style evaluation, and release-readiness analysis.
- Analyzed J2EE application behavior under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and performance-focused cutover support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OC CI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported test planning, phased validation, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and troubleshooting documentation.

Technologies: Python, SQL, Pandas, Oracle, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting and Monitoring

Built Python/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights for proactive risk mitigation.

Technologies: Python, Pandas, PySpark, SQL, Prophet, BMC TrueSight, Data Validation

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL, Data Quality

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation